

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“Jnana Sangama”, Belagavi-590014, Karnataka



INTERNSHIP REPORT ON

“SPAM MAIL DETECTION USING MACHINE LEARNING”

Submitted in partial fulfilment for the requirement of 8th semester.

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

Certificate that the INTERNSHIP on topic “**Spam Mail Detection**” on domain “Machine Learning” has been successfully submitted by **Abhilash Madhav M (1AH19CS002)** a bonafide student at **ACS College of Engineering** affiliated to **Visvesvaraya Technological University, Belagavi** during the year 2022-2023. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report submitted to the department. The Internship report has been approved as it satisfies the academic requirement in respect of the internship work as prescribed in 8th semester.

Signature of the Guide

**Signature of
Coordinator**

Signature of HOD

Name of the Examiner

Signature with Date

1. _____

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DECLARATION

I **Abhilash Madhav M(1AH19CS002)**, bonafide student of 8th semester of B.E of **ACS College of Engineering, Bangalore**, hereby declare that Internship Report entitled “**SPAM MAIL DETECTION USING MACHINE LEARNING**” has been carried out by under the guidance of Ms. Ganga BM , the work undertaken by me during the Internship of 4 weeks that I have undergone at “**PRINSTON SMART ENGINEERS Pvt. Ltd.**” during 20/08/2022 to 20/09/2022 in partial fulfilment of the requirement for the award of degree of Bachelor of Engineering in Computer Science Engineering of the Visvesvaraya Technological University, Belgaum during the academic year 2022- 2023.

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CERTIFICATE

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Abhilash Madhav M

1AH19CS002

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CHAPTER 1

INTRODUCTION

COMPANY PROFILE

Technologies head - quartered in Bangalore, India. is established by Prinston Smart Engineers having decades of experience across India & Middle East in controls and automation industry. Technologies commenced business trading as a designer and installer of multi brand controls, automation distribution systems, providing a complete range of Custom Installation services and has an extensive experience in the field of complex turn-key solutions development and integration. We propose, design, develop, train, install, integrate, operate and maintain the state-of-the-art IBMS/SCADA and the PLC automation systems and solutions. The services we offer span the full development / integration lifecycle from definition of requirements to field testing of implemented solutions.

Aim of company is to predominately provide a comprehensive training service in helping candidates to improve their skills and qualification in order to enhance their employability options. This will incorporate a high-quality intensive training to sustain in industry and to fulfill a dream of working in core industry.

The organization is operated by Asif Akhter who is Founder, Director and CEO of the company. The company initiated operations in India from 2004. But it exists in the market for almost 2 decades. They executed lots of projects in Middle East. The company offers people the opportunities to accelerate more rapidly than is possible elsewhere. They will continue to drive the greater dimension length to identify and recruit the very best person for every position. They are big enough to solve the client's problems, yet substantial enough to sustain the loyalty, intimacy and culture that they all treasure and contributes greatly to the success.

CHAPTER 2

DOMAIN INTRODUCTION

Python is a widely used, interpreted, object-oriented, and high-level programming language with dynamic semantics, used for general-purpose programming. It was created by Guido van Rossum, and first released on February 20, 1991. While you may know the python as a large snake, the name of the Python programming language comes from an old BBC television comedy sketch series called Monty Python's Flying Circus.

One of the amazing features of Python is the fact that it is actually one person's work. Usually, new programming languages are developed and published by large companies employing lots of professionals, and due to copyright rules, it is very hard to name any of the people involved in the project. Python is an exception. Of course, van Rossum did not develop and evolve all the Python components himself. The speed with which Python has spread around the world is a result of the continuous work of thousands (very often anonymous) programmers, testers, users (many of them aren't IT specialists) and enthusiasts, but it must be said that the very first idea (the seed from which Python sprouted) came to one head – Guido's.

2.1 Python goals

In 1999, Guido van Rossum defined his goals for Python:

- an easy and intuitive language just as powerful as those of the major competitors;
- open source, so anyone can contribute to its development;
- code that is as understandable as plain English;
- suitable for everyday tasks, allowing for short development times.

About 20 years later, it is clear that all these intentions have been fulfilled. Some sources say that Python is the third-most popular programming language in the world, while others claim it's the fifth. Either way, it still occupies a high rank in the top ten of the TIOBE Programming Community and PYPL Popularity of Programming Language Indexes

Python isn't a young language. It is mature and trustworthy. It's not a one-hit wonder. It's a bright star in the programming firmament, and time spent learning Python is a very good investment.

2.2 Why Python?

What makes Python so special? How does it happen that programmers, young and old, experienced and novice, want to use it? How did it happen that large companies adopted Python and implemented their flagship products using it?

There are many reasons – we've listed some of them already, but let's enumerate them again in a more practical manner:

- it's easy to learn – the time needed to learn Python is shorter than for many other languages; this means that it's possible to start the actual programming faster;
- it's easy to use for writing new software – it's often possible to write code faster when using Python;
- it's easy to understand – it's also often easier to understand someone else's code faster if it is written in Python;
- it's easy to obtain, install and deploy – Python is free, open and multiplatform; not all languages can boast that.

2.3 Python in Action

It's used extensively to implement complex Internet services like search engines, cloud storage and tools, social media and so on. Whenever you use any of these services, you are actually very close to Python, although you wouldn't know it. Many developing tools are implemented in Python.

More and more everyday use applications are being written in Python. Lots of scientists have abandoned expensive proprietary tools and switched to Python. Lots of IT project testers have started using Python to carry out repeatable test procedures. The list is long.

2.4 Python Examples

- Web and Internet development (e.g., Django and Pyramid frameworks, Flask and Bottle micro-frameworks)
- Scientific and numeric computing (e.g., SciPy – a collection of packages for the purposes of mathematics, science, and engineering; IPython – an interactive shell that features editing and recording of work sessions)• Education (it's a brilliant language for teaching programming!)
- Desktop GUIs (e.g., wxWidgets, Kivy, Qt)
- Software Development (build control, management, and testing – Scons, Buildbot, ApacheGump, Roundup, Trac)
- Business applications (ERP and e-commerce systems – Odoo, Tryton)
- Games (e.g., Battlefield series, Sid Meier's Civilization IV...), websites and services (e.g., Dropbox, UBER, Pinterest, BuzzFeed...).

CHAPTER 3**PROJECT INTRODUCTION**

Spam email is unsolicited and unwanted junk email sent out in massive amount or in bulk to an indiscriminate recipient list. Generally, spam is sent for commercial purposes. It is sent in massive volume by botnets, networks of infected computers. Spam email can often be a malicious attempt to gain access to your system. Spam prevents the user from making full and good utilization of cpu time, storage capacity and network bandwidth. It becomes a huge problem especially at times when there are Spam mails which come in between important business mails. Hence, it becomes inevitable to solve such problems which are encountered by spam email. So, this problem can be solved by using Machine Learning methods which can successfully detect and filter spam. It is also important to find out which technique or algorithm can best fit in the purpose of classifying spam mail.

The person responsible for sending the spam messages is referred to as the spammer. Such a person gathers email addresses from different websites, chat rooms etc. The huge volume of Spam mails flowing through the computer networks have destructive effects on the memory space of the email server, communication bandwidth, cpu power and user time. In all, existing system does not find spam mails effectively. Hence, it also results in untold financial losses to many users. It leads to low test and prediction accuracy, less security and also loss of data.

CHAPTER 4

LIBRARIES

4.1 PANDAS

Then Pandas is the library you are looking for. It is easily one of the most sought after libraries for python, and it has a relatively easy learning curve. Pandas play an important role in Data Science. This Python pandas tutorial helps you to build skills for data scientist and data analyst. Data is an integral part of our current world. It helps us predict various events and gives a certain direction to our lives. Pandas help us control and manipulate such data.

Thus without a grasp over the knowledge of Pandas, you can completely forget about trying to become a Data Scientist or Data Analyst. Pandas are an essential tool for a beginners journey to work with data. Pandas provide essential data structures like series, data frames, and panels which help in manipulating data sets and time series. It is free to use and an open source library, making it one of the most widely used data science libraries in the world. Pandas possess the power to perform various tasks.

Whether it is computing tasks like finding the mean, median and mode of data, or a task of handling large CSV files and manipulating the contents according to our will, Pandas can do it all.

Key Components of Pandas

Mainly, a Pandas Data Frame can be compared to a two-dimensional array. These are heavily used to store and manipulate data

- Pandas Series- A series in Pandas can be thought of as a unidimensional array that is used to handle and manipulate data which is stored in it.
- Pandas Data Frame- This is a data structure in Pandas, which is made up of multiple series.

Pandas Library Architecture

This Python Pandas Tutorial is incomplete without library architecture. So, let's discuss the

file hierarchy in pandas.

- pandas/core: Consists of data structures about the Pandas library.
- pandas/src: Holds the basic functionality of Pandas depend on certain algorithms. They are usually written in C or Cython.
- pandas/io: Carries the tools to input and output, files, data, etc
- pandas/tools: Codes and algorithms for various functions and operations in Pandas. For example: Merge and join, concatenation, etc.
- pandas/sparse: Carries the sparse versions, i.e., the versions made to handle missing values of various Data Structures in Pandas. • pandas/stats: Contains functions related to statistics, like linear regression
- pandas/util: Consist of testing tools and various other utilities to debug the library.
- pandas/rpy: Consists of an interface which helps to connect to R. It is called R2Py

Python Pandas Features

1. Handling of data: The Pandas library provides a really fast and efficient way to manage and explore data. It does that by providing us with Series and Data Frames, which help us not only to represent data efficiently but also manipulate it in various ways. These features of Pandas is exactly what makes it such an attractive library for data scientists.
2. Alignment and indexing: Having data is useless if you don't know where it belongs and what it tells us about. Therefore, labeling of data is of utmost importance. Another important factor is an organization, without which data would be impossible to read. These two needs: Organization and labeling of data are perfectly taken care of by the intelligent methods of alignment and indexing, which can be found within Pandas.
3. Handling missing data: As discussed above, data can be quite confusing to read. But that is not even one of the major problems. Data is very crude in nature and one of the many problems associated with data is the occurrence of missing data or value. Therefore, it is pertinent to handle the missing values properly so that they do not adulterate our study results. Some Pandas features have you covered on this end because handling missing values is integrated within the library.
4. Cleaning up data: Like we just said, Data can be very crude. Therefore it is really messy, so much so that performing any analysis over such data would lead to severely wrong results. Thus it is of extreme importance that we clean our data up, and this Pandas feature is easily provided. They

help a lot to not only make the code clean but also tidies up the data so that even the normal eye can decipher parts of the data. The cleaner the data, the better the result.

5. Input and output tools: Pandas provide a wide array of built-in tools for the purpose of reading and writing data. While analysing you will obviously need to read and write data into data structures, web service, databases, etc. This has been made extremely simple with the help of Pandas' inbuilt tools. In other languages, it would probably take a lot of code to generate the same results, which would only slow down the process of analysing.
6. Multiple file formats supported: Data these days can be found in so many different file formats, that it becomes crucial that libraries used for data analysis can read various file formats. Pandas covers this sector with a huge scope of file formats supported. Whether it is a JSON or CSV, Pandas can support it all, including Excel and HDF5.
7. Merging and joining of datasets: While analysing data we constantly need to merge and join multiple datasets to create a final dataset to be able to properly analyse it. This is important because if the datasets aren't merged or joined properly, then it is going to affect the results adversely and we do not want that. Pandas can help to merge various datasets, with extreme efficiency so that we don't face any problems while analysing the data.
8. A lot of time series: These Pandas features won't make sense to beginners right away, but they will be of great use in the future. These features include the likes of moving window statistics and frequency conversion. So, as we go deeper into learning Pandas we will see how essential and useful these features are, for a data scientist.
9. Optimized performance: Pandas is said to have a really optimized performance, which makes it really fast and suitable for data science. The critical code for Pandas is written in C or Cython, which makes it extremely responsive and fast.
10. Python support: This feature of Pandas is the deal closer. With an insane amount of helpful libraries at your disposal Python has become one of the most sought after programming languages for data analysis. Thus Pandas being a part of Python and allowing us to access the other libraries like NumPy and MatPlot Lib.
11. Visualize: Visualizing the data is an important part of data science. It is what makes the results of the study understandable by human eyes. Pandas have an in-built ability to help you plot your data and see the various kinds of graphs formed. Without visualization, data analysis would make no sense to most of the population.
12. Grouping: Having the ability to separate your data and grouping it according to the criteria you want, is pretty essential. With the help of the features of Pandas like GroupBy, you can split data

into categories of your choice, according to the criteria you set. The GroupBy function splits the data, implements a function and then combines the results.

13. Mask data: Sometimes, certain data is not needed for analysis of data and thus it is important that you filter your data according to the things you want from it. Using the mask function in Pandas allows you exactly to do that. It is extremely useful since whenever it finds data which meets the criteria you set for elimination, it turns the data into a missing value.
14. Unique data: Data always has a lot of repetition, therefore it is important that you are able to analyze data which has only unique values. This is present in the Python Pandas features and lets the user see the unique values in the dataset with the function `dataset.column.unique()`. Where “dataset” and “column” are the names of your dataset and column, respectively.
15. Perform mathematical operations on the data: The apply function in Pandas allows you to implement a mathematical operation on the data. This helps enormously, because sometimes the dataset you have, is just not of the correct order. This will be correct by simply using a mathematical operation on the dataset. This is one of the most attractive features of Pandas.

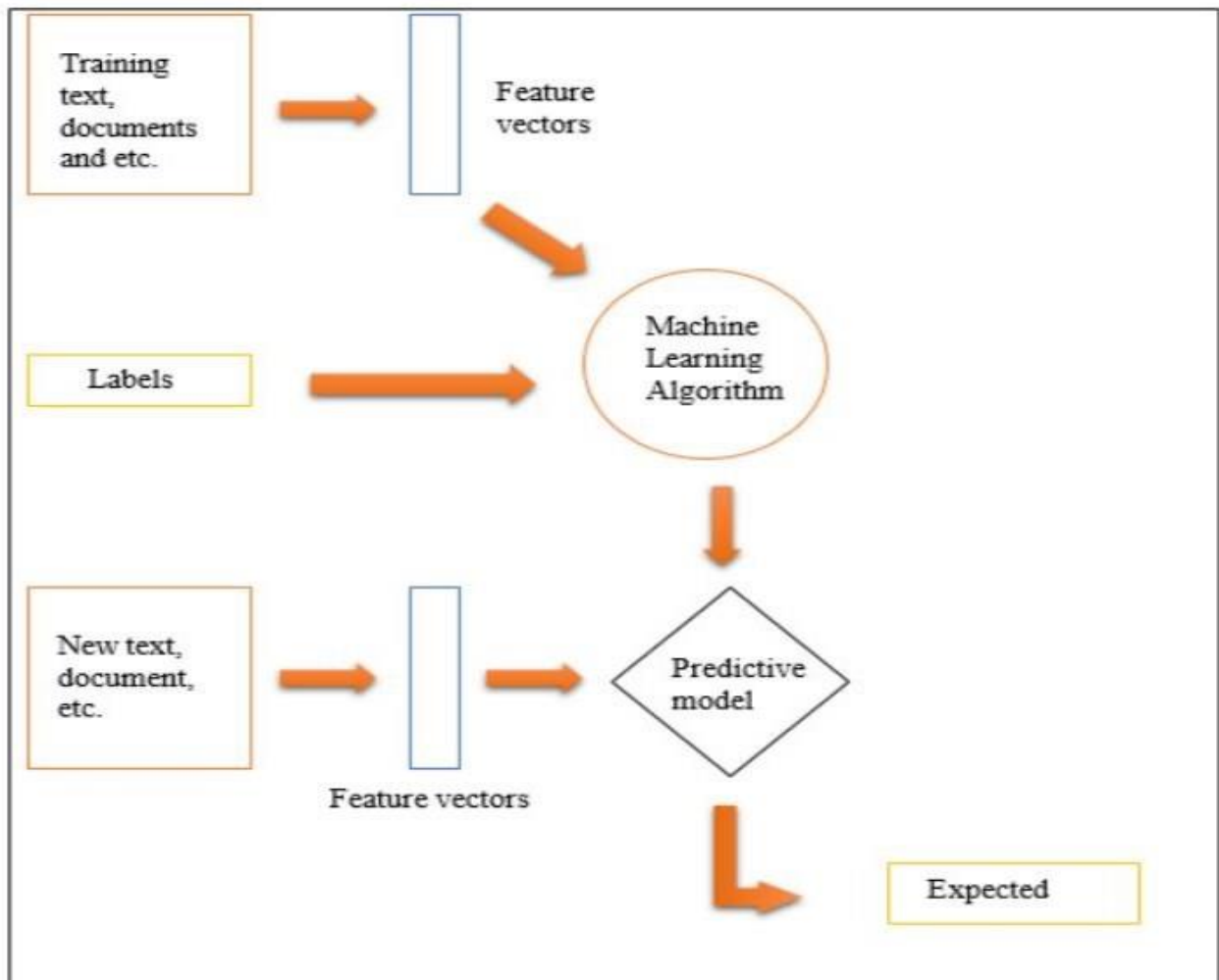
4.2 SCIKIT LEARN

Scikit-learn (Sklearn) is the most useful and robust library for machine learning in Python. It provides a selection of efficient tools for machine learning and statistical modeling including classification, regression, clustering and dimensionality reduction via a consistent interface in Python. This library, which is largely written in Python, is built upon NumPy, SciPy and Matplotlib.

As a high-level library, it lets you define a predictive data model in just a few lines of code, and then use that model to fit your data. It's versatile and integrates well with other Python libraries, such as matplotlib for plotting, numpy for array vectorization, and pandas for dataframes.

CHAPTER 5

IMPLEMENTATION



SPAM DETECTION MODEL FLOW

Code :

Importing the Dependencies"""

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
```

"""Data Collection & Pre-Processing"""

```
# @title# loading the data from csv file to a pandas Dataframe
raw_mail_data = pd.read_csv('/content/mail_data.csv')
print(raw_mail_data)
```

```

# replace the null values with a null stringmail_data =
raw_mail_data.where((pd.notnull(raw_mail_data)),")

# printing the first 5 rows of the dataframemail_data.head()

# checking the number of rows and columns in the dataframemail_data.shape

"""Label Encoding"""

# label spam mail as 0; ham mail as 1;

mail_data.loc[mail_data['Category'] == 'spam', 'Category',] = 0mail_data.loc[mail_data['Category']
== 'ham', 'Category',] = 1

"""spam - 0

ham - 1"""

# separating the data as texts and label

X = mail_data['Message']

Y = mail_data['Category']

print(X)

print(Y)

"""Splitting the data into training data & test data"""

X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=3

print(X.shape)print(X_train.shape)print(X_test.shape)

"""Feature Extraction"""

# transform the text data to feature vectors that can be used as input to the Logistic regression

feature_extraction = TfidfVectorizer(min_df = 1, stop_words='english', lowercase=True)

X_train_features = feature_extraction.fit_transform(X_train)X_test_features =
feature_extraction.transform(X_test)

```

```

)

# convert Y_train and Y_test values as integers

Y_train = Y_train.astype('int')Y_test = Y_test.astype('int')

print(X_train)

print(X_train_features)

"""Training the Model

Logistic Regression"""

model = LogisticRegression()

# training the Logistic Regression model with the training data
model.fit(X_train_features, Y_train)

"""Evaluating the trained model"""

# prediction on training data

prediction_on_training_data = model.predict(X_train_features)accuracy_on_training_data =
accuracy_score(Y_train, prediction_on_training_data)

print('Accuracy on training data : ', accuracy_on_training_data)

# prediction on test data

prediction_on_test_data = model.predict(X_test_features)accuracy_on_test_data =
accuracy_score(Y_test, prediction_on_test_data)

print('Accuracy on test data : ', accuracy_on_test_data)

"""Building a Predictive System"""

input_mail = ['I've been searching for the right words to thank you for this breather. I promise i
wont take your help for granted and will fulfil my promise. You have been wonderful and a blessing
at all times"]

# convert text to feature vectorsinput_data_features = feature_extraction.transform(input_mail)

# making prediction
prediction = model.predict(input_data_features)print(prediction)

if (prediction[0]==1):
    print('Ham mail')
else:print('Spam mail')

```

CHAPTER 6

RESULTS

1) Loading the mail data from csv file to a pandas Dataframe

```
#@title loading the data from csv file to a pandas Dataframe
#
raw_mail_data = pd.read_csv('/content/mail_data.csv')

[ ] print(raw_mail_data)
```

	Category	Message
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...
...	...	...
5567	spam	This is the 2nd time we have tried 2 contact u...
5568	ham	Will ü b going to esplanade fr home?
5569	ham	Pity, * was in mood for that. So...any other s...
5570	ham	The guy did some bitching but I acted like i'd...
5571	ham	Rofl. Its true to its name

[5572 rows x 2 columns]

2) Replacing the null values with a null string and printing the first 5 rows of the mail_data dataframe.

```
# printing the first 5 rows of the dataframe
mail_data.head()
```

	Category	Message
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...

SPAM MAIL DETECTION

3) Labelling spam mail as 0 and ham mail as 1.

```
Label Encoding

[7] # label spam mail as 0; ham mail as 1;

mail_data.loc[mail_data['Category'] == 'spam', 'Category',] = 0
mail_data.loc[mail_data['Category'] == 'ham', 'Category',] = 1

spam - 0
ham - 1
```

4) Evaluating the trained model

```
Evaluating the trained model

[18] # prediction on training data

prediction_on_training_data = model.predict(X_train_features)
accuracy_on_training_data = accuracy_score(Y_train, prediction_on_training_data)

[19] print('Accuracy on training data : ', accuracy_on_training_data)

Accuracy on training data : 0.9670181736594121

[20] # prediction on test data

prediction_on_test_data = model.predict(X_test_features)
accuracy_on_test_data = accuracy_score(Y_test, prediction_on_test_data)

[21] print('Accuracy on test data : ', accuracy_on_test_data)

Accuracy on test data : 0.9659192825112107
```

SPAM MAIL DETECTION

4) FINAL RESULT : Building a Predictive System that identifies a given mail as 'ham or spam'.

```
✓ [22] input_mail = ["I've been searching for the right words to thank you for this breather.  
0s  
  
# convert text to feature vectors  
input_data_features = feature_extraction.transform(input_mail)  
  
# making prediction  
  
prediction = model.predict(input_data_features)  
print(prediction)  
  
if (prediction[0]==1):  
    print('Ham mail')  
  
else:  
    print('Spam mail')  
  
[1]  
Ham mail
```

CHAPTER 7**CONCLUSION**

The spam mail dataset consists of two data forms: train and test. The dataset contains a randomly selected collection of emails in plain text format, which have been labelled as HAM or SPAM. The training data is used to build a model for classifying emails into HAM and SPAM. The test data is used to check the accuracy of the model built with the training data. The emails in the learning data are in plain text format. We need to convert the plain text into features that can represent the emails. Using these features we can then use a learning algorithm on the emails. A number of pre-processing steps are first performed. Finally once the data model has been trained with sufficient training data and evaluated with test data for checking accuracy, a random mail from the data set is taken as input for the model. The model then classifies the data thereby predicting whether the input mail is a 'spam' or a 'ham'.

CHAPTER 8

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